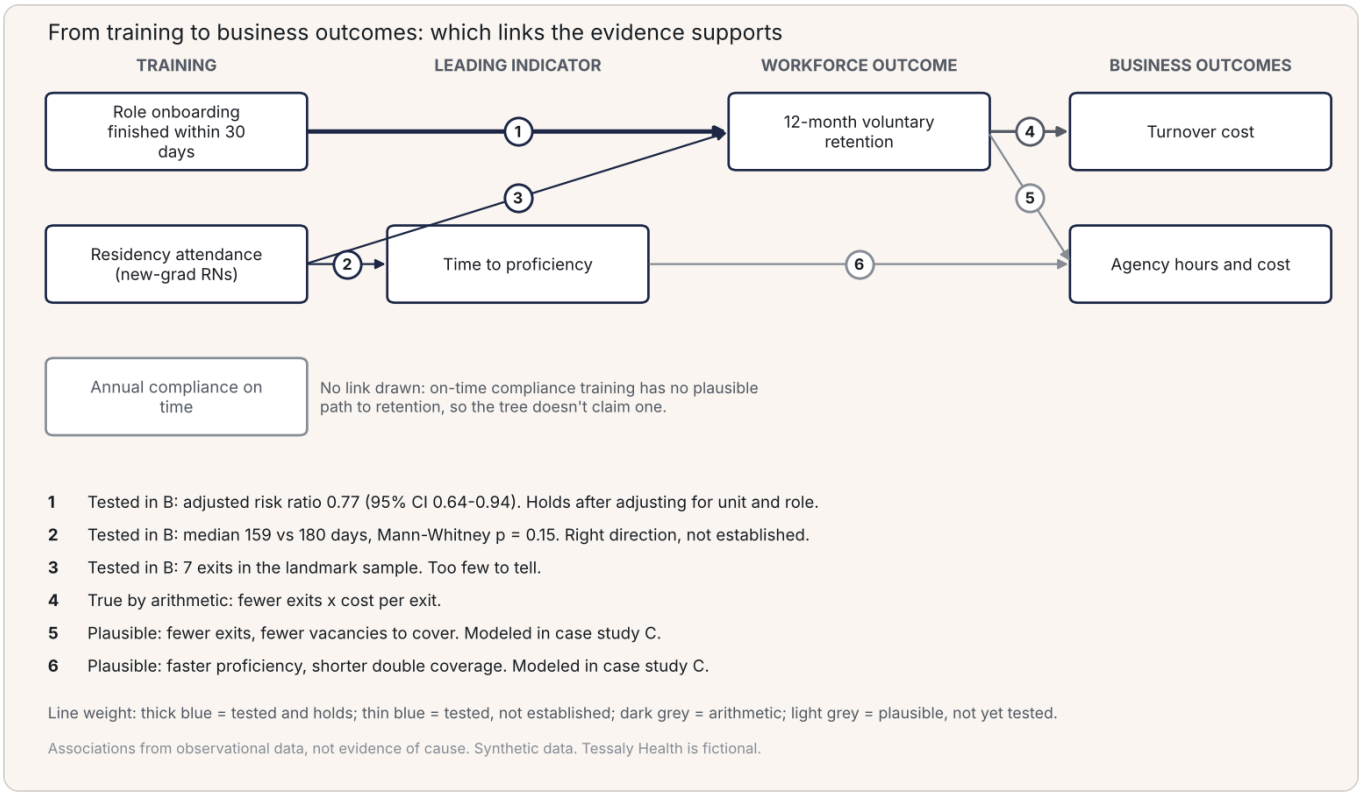
Matthew McKay, Ph.D.

Workforce analytics metric tree: training to retention (LIB-05)

Purpose: a one-page map, for learning and HR leaders, of how training is supposed to reach business outcomes, and how strong the evidence is for each link. Use it before anyone claims a training ROI.

Worked example at Tessaly Health, a fictional health system, using synthetic data from case study B.



How to read and use it

Link strength	Meaning	What you can say
Tested and holds	Association found, survives adjustment and sensitivity checks	“Linked to”; size it with the interval, and don’t call it cause
Tested, not established	Right direction, interval includes no effect	“We can’t tell yet”; say why (usually sample size)
Arithmetic	True by definition once the inputs are known	State the formula and the assumed inputs
Plausible, not yet tested	A mechanism exists, no evidence here	Name it as a hypothesis, and plan the test
No link	No credible mechanism	Don’t draw it, even if a correlation turns up

To build one for another program: list the training exposures on the left and the business outcomes on the right. Draw only the links with a mechanism you can say in one sentence. Then label each by the evidence you actually have, not the evidence you expect.

Why it’s built this way

Training ROI claims usually fail at a missing middle link: a course is credited with a cost saving that depends on two untested steps. Drawing the whole chain makes those steps visible, and labeling each link’s evidence keeps a strong result on one link from lending credibility to the rest. I left compliance training unconnected on purpose. Compliance modules exist to meet obligations, not to retain people, and drawing a line there would invite someone to find a spurious correlation and report it. The tree also tells case study C which links it has to model rather than assume.

Format: image (PNG and SVG in `outputs/`), with this page.

Matthew McKay, Ph.D.

Personal project. All data is synthetic or public, and the company is fictional. Views are my own and not affiliated with or endorsed by any current or former employer.